CS2320 — Introduction to Graph Theory

Tier	B
Phase (order)	4
Do this first	CS1202
Estimated effort	10 weeks
Unlocks	CS6130, CS6240, CS2800 graph algorithms

Roadmap

1. Follow Bondy-Murty chapters in order: basics, trees, connectivity, matchings, planarity, colouring, digraphs.
2. Write every proof out; this course trains proof writing.

Video courses (all links checked on YouTube on 2026-10-22)

🎓 Main course: follow it start to end

- Computer - Graph Theory (40 videos). NPTEL *Graph Theory* (nptelhrd).

🎥 Other full lecture series (pick one if the main one doesn't click, or for topics it skips)

- Graph Theory part-1 (9 videos). Sarada Herke *Graph Theory* (part 1 of her series; parts 2-10 are on her channel): proof-focused, matches Bondy-Murty.

⚡ Short videos for intuition and revision

- Graph Theory Playlist (43 videos). WilliamFiset *Graph Theory*: the algorithmic side, with code.
- Network Flow playlist (11 videos). WilliamFiset *Network Flow*.

Study plan (week by week)

Lecture numbers are positions in the main playlist above. Click a number to open that video.

Week	Lectures	What to do
1-2	1–8	Vertex cover, matchings, Konig, Hall, Tutte, Gallai-Milgram, Dilworth. Implement bipartite matching (Hopcroft-Karp).
3	9–12	Connectivity, Menger's theorem, minors.
4-5	13–22	Vertex/edge colouring, Brooks, Vizing, planarity, Kuratowski, list colouring, chromatic polynomials.
6	23–30	Perfect graphs, chordal/interval graphs, Hamiltonicity.

Week	Lectures	What to do
7	31–34	Network flows, max-flow min-cut. Implement Edmonds-Karp.
8	35–40	Probabilistic method, random graphs, graph minors.

▼ **All lectures in *Computer - Graph Theory* (40 videos)**

1. Mod-01 Lec-01 Introduction: Vertex cover and independent set
2. Mod-01 Lec-02 Matchings: Konig's theorem and Hall's theorem
3. Mod-01 Lec-03 More on Hall's theorem and some applications
4. Mod-01 Lec-04 Tutte's theorem on existence of a perfect matching
5. Mod-01 Lec-05 More on Tutte's theorem
6. Mod-01 Lec-06 More on Matchings
7. Mod-01 Lec-07 Dominating set, path cover
8. Mod-01 Lec-08 Gallai -- Millgram theorem, Dilworth's theorem
9. Mod-02 Lec-09 Connectivity: 2-connected and 3- connected graphs
10. Mod-02 Lec-10 Menger's theorem
11. Mod-02 Lec-11 More on connectivity: k- linkedness
12. Mod-02 Lec-12 Minors, topological minors and more on k- linkedness
13. Mod-03 Lec-13 Vertex coloring: Brooks theorem
14. Mod-03 Lec-14 More on vertex coloring
15. Mod-03 Lec-15 Edge coloring: Vizing's theorem
16. Mod-03 Lec-16 Proof of Vizing's theorem, Introduction to planarity
17. Mod-03 Lec-17 5- coloring planar graphs, Kuratowsky's theorem
18. Mod-03 Lec-18 Proof of Kuratowsky's theorem, List coloring
19. Mod-03 Lec-19 List chromatic index
20. Mod-03 Lec-20 Adjacency polynomial of a graph and combinatorial Nullstellensatz
21. Mod-03 Lec-21 Chromatic polynomial, k - critical graphs
22. Mod-03 Lec-22 Gallai-Roy theorem, Acyclic coloring, Hadwiger's conjecture
23. Mod-04 Lec-23 Perfect graphs: Examples
24. Mod-04 Lec-24 Interval graphs, chordal graphs
25. Mod-04 Lec-25 Proof of weak perfect graph theorem (WPGT)
26. Mod-04 Lec-26 Second proof of WPGT, Some non-perfect graph classes
27. Mod-04 Lec-27 More special classes of graphs
28. Mod-04 Lec-28 Boxicity,Sphericity, Hamiltonian circuits
29. Mod-04 Lec-29 More on Hamiltonicity: Chvatal's theorem
30. Mod-04 Lec-30 Chvatal's theorem, toughness, Hamiltonicity and 4-color conjecture
31. Mod-05 Lec-31 Network flows: Max flow mincut theorem
32. Mod-05 Lec-32 More on network flows: Circulations
33. Mod-05 Lec-33 Circulations and tensions
34. Mod-05 Lec-34 More on circulations and tensions, flow number and Tutte's flow conjectures
35. Mod-06 Lec-35 Random graphs and probabilistic method: Preliminaries

36. Mod-06 Lec-36 Probabilistic method: Markov's inequality, Ramsey number
37. Mod-06 Lec-37 Probabilistic method: Graphs of high girth and high chromatic number
38. Mod-06 Lec-38 Probabilistic method: Second moment method, Lovasz local lemma
39. Mod-07 Lec-39 Graph minors and Hadwiger's conjecture
40. Mod-07 Lec-40 More on graph minors, tree decompositions

Books, notes and websites

- Book: Bondy & Murty, *Graph Theory* (the course text)
- Diestel, *Graph Theory* — diestel-graph-theory.com (free electronic edition)
- ✅ NPTEL *Graph Theory* (L. Sunil Chandran, IISc) — nptel.ac.in/courses/106108054 (nearest full-course match; uses Diestel)
- William Fiset's graph theory videos for algorithmic intuition

Practice / proof of understanding

Prove Cayley's formula, Hall's theorem and the 5-colour theorem; code a bipartite matching.

What your IITM course sheet says

From the PDF: CS2320

Official description: The goal of this course is to introduce students to Graph Theory (a topic that finds many applications in various areas of computer science and other fields such as electrical engineering). Apart from learning different graph theoretical concepts, the students will also receive further training in writing rigorous proofs. Several of these proofs are constructive, and thus intrinsically related to algorithms.

Official syllabus (trimmed):

> I. Basics (6 lectures): graphs and subgraphs; isomorphisms and automorphisms; representations of graphs; degrees and degree sequences; walks, trails, paths and cycles; connected graphs; Eulerian and Hamilton graphs. II. Trees and bipartite graphs (5 lectures): characterizations and properties; spanning trees; Cay- ley's formula. III. Connectivity (7 lectures): edge connectivity, vertex connectivity, ear decompositions, block decomposition, Menger's Theorems. IV. Matchings and Covers (6 lectures): matchings and augmenting paths, Hall's Theorem, Konig's Theorem, Tutte's Theorem (without proof). V. Planarity (5 lectures): Euler's Formula and its consequences, Planar duality, Kuratowski's Theorem (without proof). VI. Colorings (7 lectures): Edge-colorings of bipartite graphs; Vizing's Theorem (without proof); coloring number and degeneracy; chromatic number and clique number; Brooks' Theorem; Four color problem (with brief history); 6-color and 5-color theorems. VII. Directed graphs (4 lectures): Basics, DAGs, tournaments.

Official textbooks: Graph Theory by J. A. Bondy and U. S. R. Murty (Springer 2008)

Official references: 1. Graph Theory with Applications by J. A. Bondy and U. S. R. Murty (Macmillan 1976) 2. Introduction to Graph Theory by D. B. West (Prentice Hall 1999)